Bank Application System

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INTRODUCTION

- The system allows users to create and manage bank accounts efficiently. Users can perform transactions such as deposit, withdrawal, and fund transfer.
- It maintains a structured database to store account, transaction, and loan details.
  This ensures secure data storage and proper record management.
- The application provides OTP-based authentication for enhanced security. This prevents unauthorized access and protects sensitive user information.
- The system also includes loan management and EMI calculation features. It helps users track loan status, repayment details, and interest calculation.

OBJECTIVES

- To implement Object-Oriented Programming concepts in a real-world application.
  The project uses classes and methods to manage accounts and loans efficiently.
- To integrate Python with MySQL database.
  This ensures proper data storage, retrieval, and transaction management.
- To simulate real banking operations.
  The system performs financial calculations like interest and EMI.

TECHNOLOGIES USED

- Python Programming Language :
  Python is used to build the core logic and implement OOP concepts.
  It handles user input, calculations, and database interaction.
- MySQL Database :
  MySQL stores account details, transactions, and loan records securely.
  It ensures data integrity using primary and foreign keys.

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DATABASE DESIGN

- Accounts Table
  Stores account number, name, balance, PIN, account type, and interest rate.
  Account number acts as the primary key for unique identification.
- Transactions Table
  Records all deposits, withdrawals, transfers, and loan repayments.
  It maintains transaction history with timestamps.
- Loans Table
  Stores loan type, loan amount, tenure, interest rate, and status.
  It tracks remaining amount until the loan is fully paid.

ACCOUNT FEATURES

- Registration with OTP
  Users must verify their identity using a one-time password.
  This adds an extra security layer during account creation.
- Deposit and Withdrawal
  Users can deposit or withdraw money securely.
  A penalty is applied if the balance falls below the minimum limit.
- Money Transfer
  Users can transfer funds between accounts.
  The system updates both sender and receiver balances automatically.

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LOAN AND EMI SYSTEM

- Loan Application
  Users can apply for Home, Gold, or Personal loans.
  Each loan type has a predefined interest rate.
- EMI Calculation
  The system calculates EMI using the standard mathematical formula.
  EMI depends on loan amount, interest rate, and tenure.
- Loan Repayment
  EMI is deducted from the account balance.
  Loan status changes to “Paid” when the amount becomes zero.

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ADDITIONAL FUNCTIONALITIES

- Calculation
  Savings account holders receive 3.2% annual interest.
  The interest amount is automatically added to the balance
- Transaction Visualization
  A pie chart displays transaction distribution.
  It helps users analyze spending and deposits visually.
- Passbook Generation
  A text file is created automatically for each account.
  It records complete transaction history with timestamps.

ADVANTAGES AND LIMITATIONS

- Advantages:
  The system ensures secure OTP authentication.
  It maintains accurate transaction records and loan tracking
- Limitations:
  The application is console-based and lacks a graphical interface.
  OTP is simulated and not sent via real SMS service.



CONCLUSION



- The Banking Management System successfully demonstrates database integration and OOP implementation.
- It performs real-world banking operations like transactions, loans, and interest calculation.
- The project shows practical understanding of financial logic and secure system design.

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Thank You